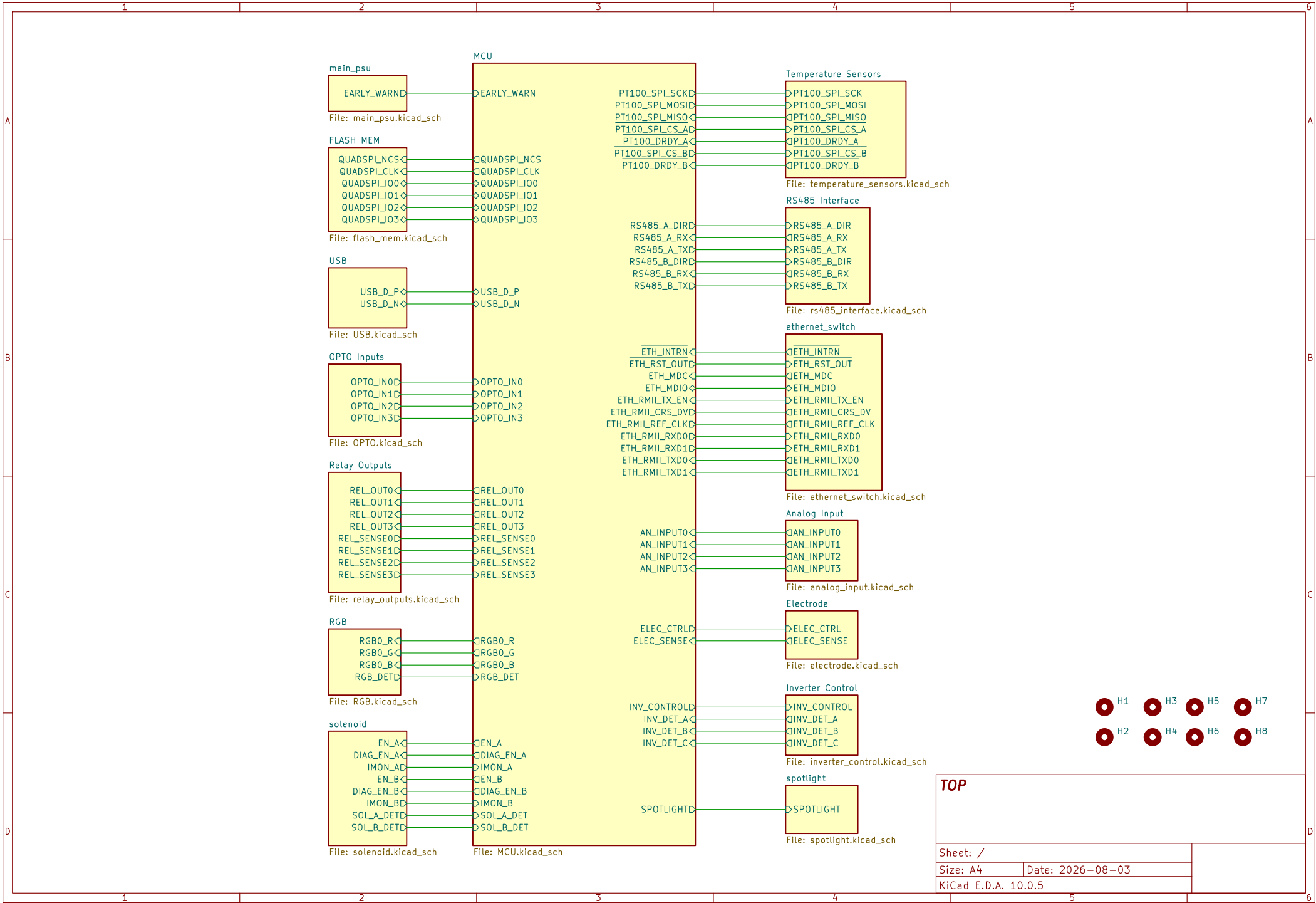
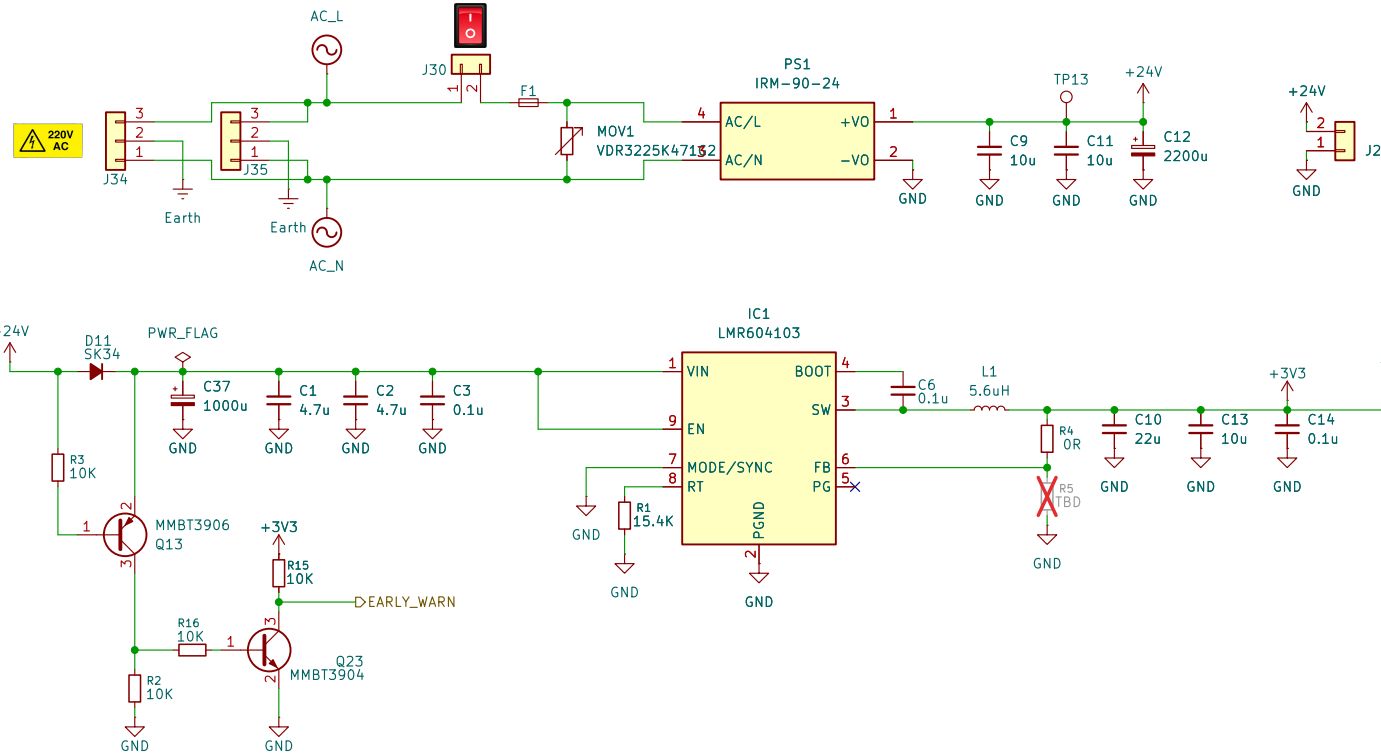
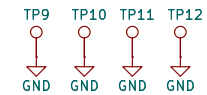
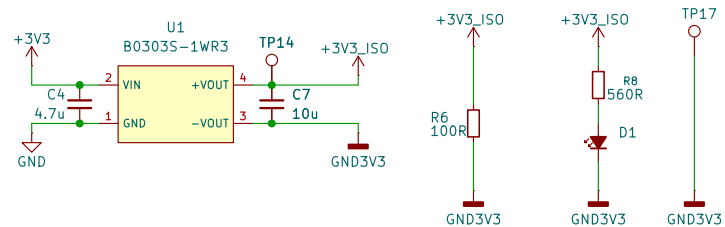






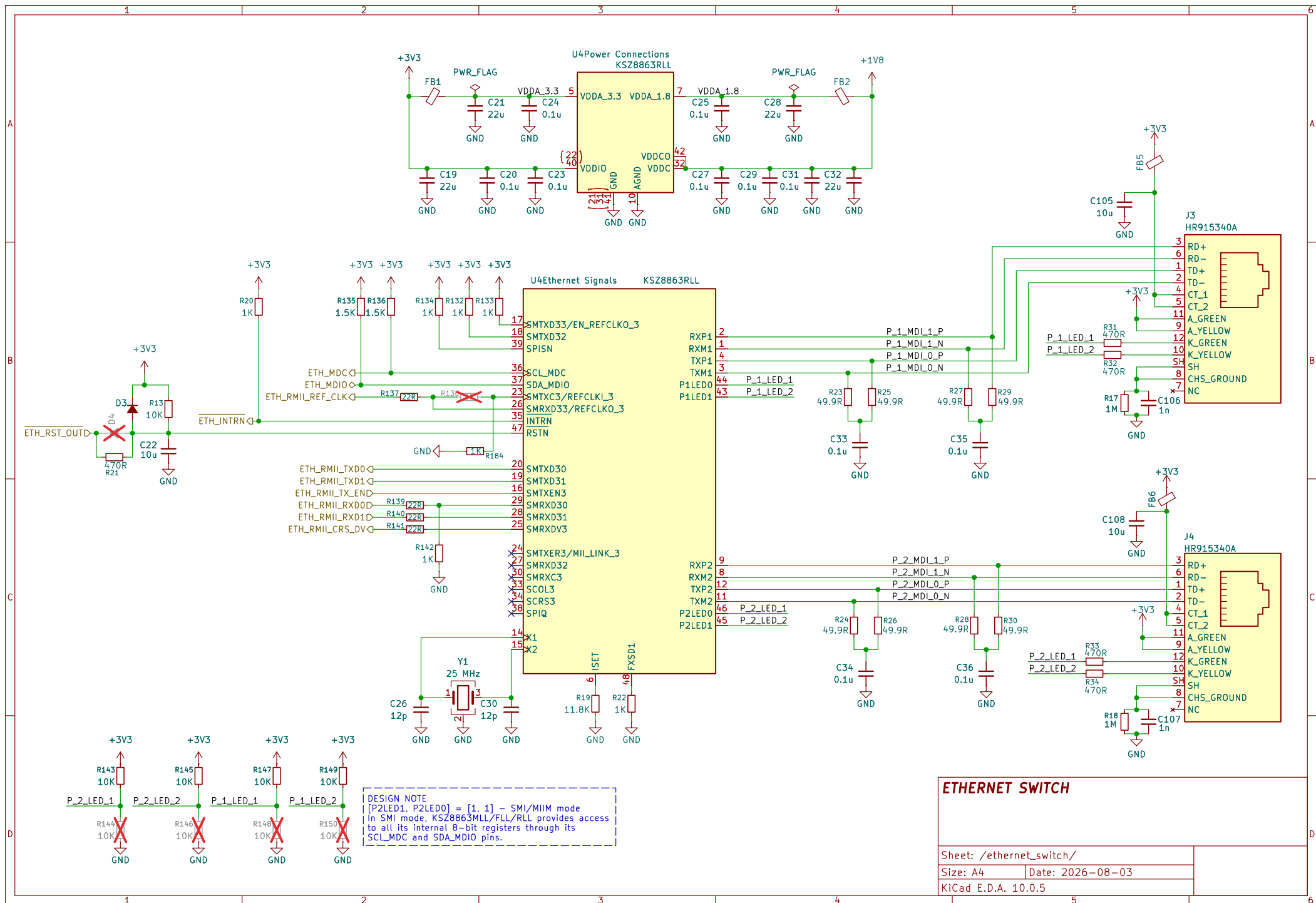
DESIGN NOTE: B0303S-1WR3 ISOLATED DC/DC

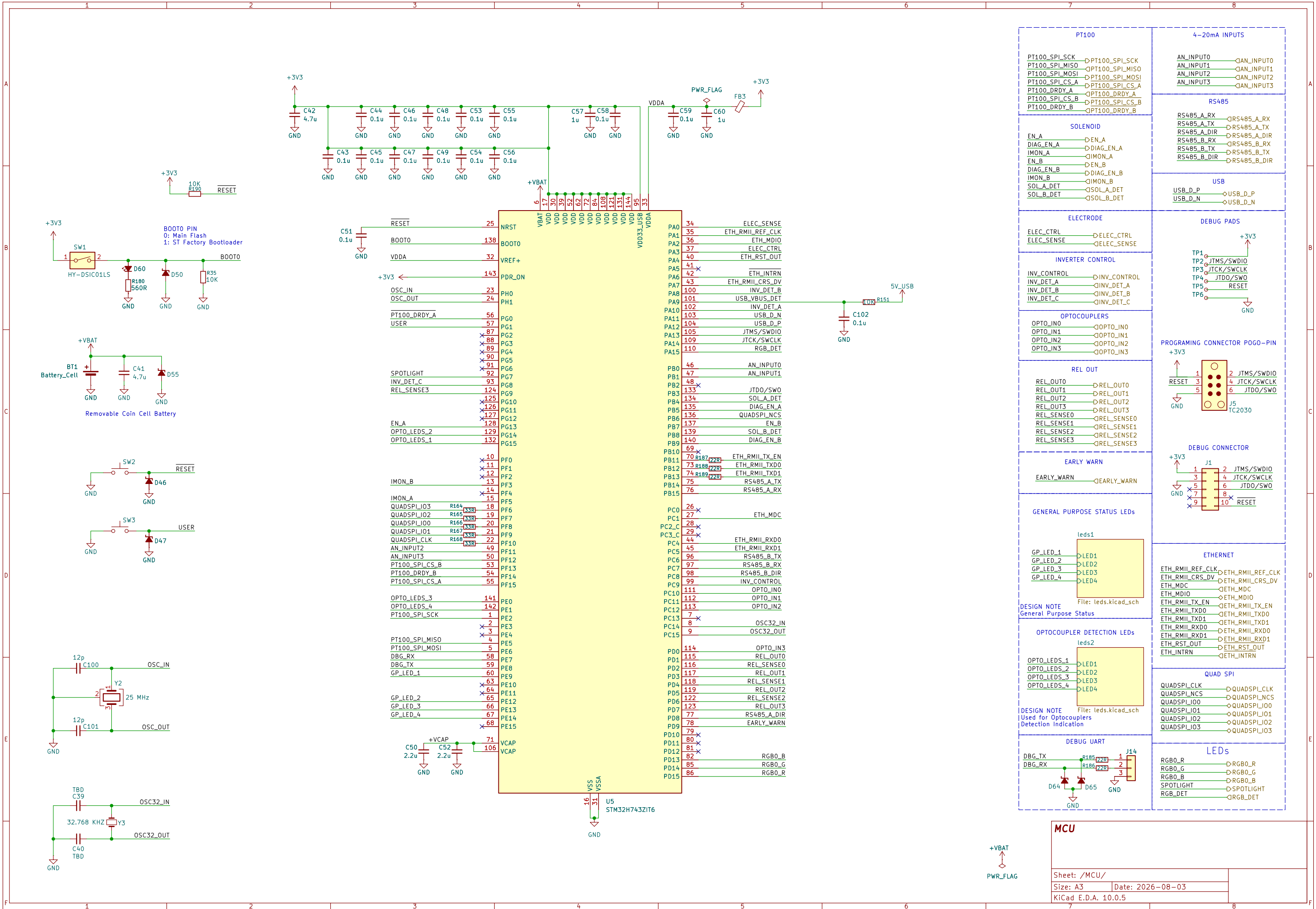
- Unregulated converter: Requires minimum 10% load (30mA) to maintain stable 3.3V_ISO output.
- Dummy Load (R85): 100Ω, 1/4W minimum rating (draws ~33mA).
- Power LED (D34 + R143): Draws ~2.7mA.
- Total Constant Load: ~35.7mA.

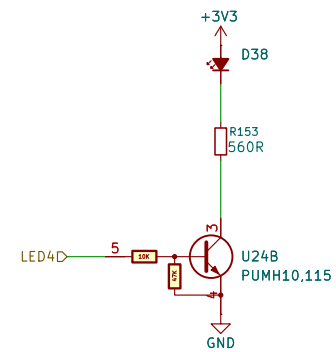
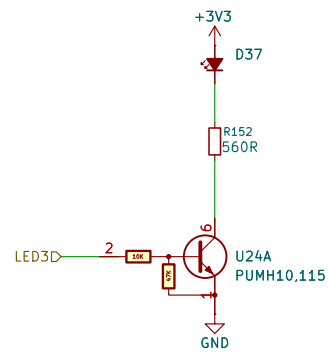
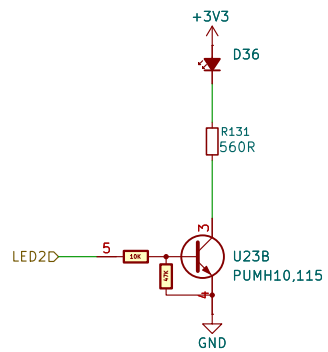
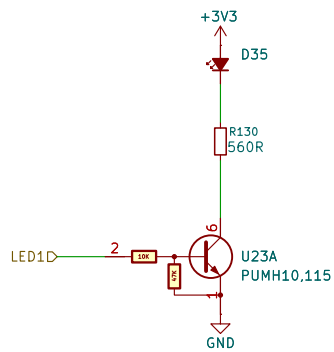


POWER SUPPLIES

Sheet: /main_psu/
Size: A4 Date: 2026-08-03
KiCad E.D.A. 10.0.5





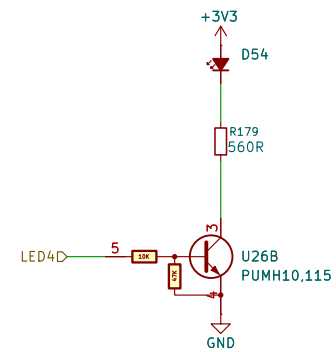
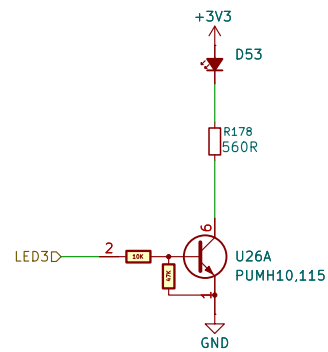
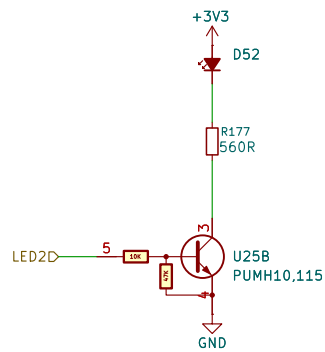
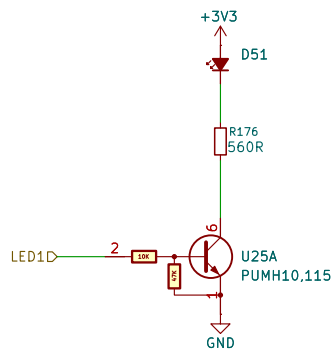


LEDs

Sheet: /MCU/leds1/

Size: A4 Date: 2026-08-03

KiCad E.D.A. 10.0.5

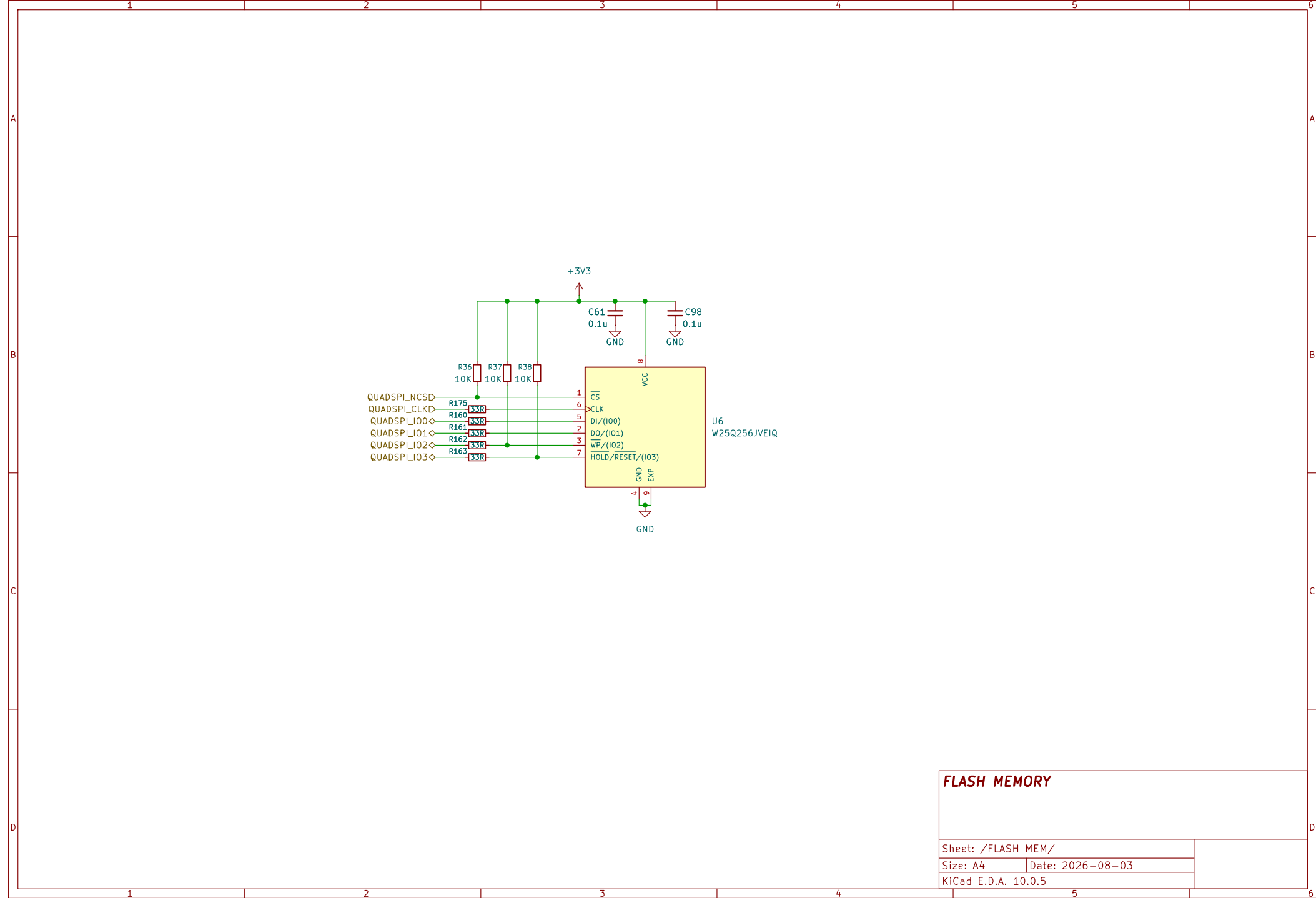


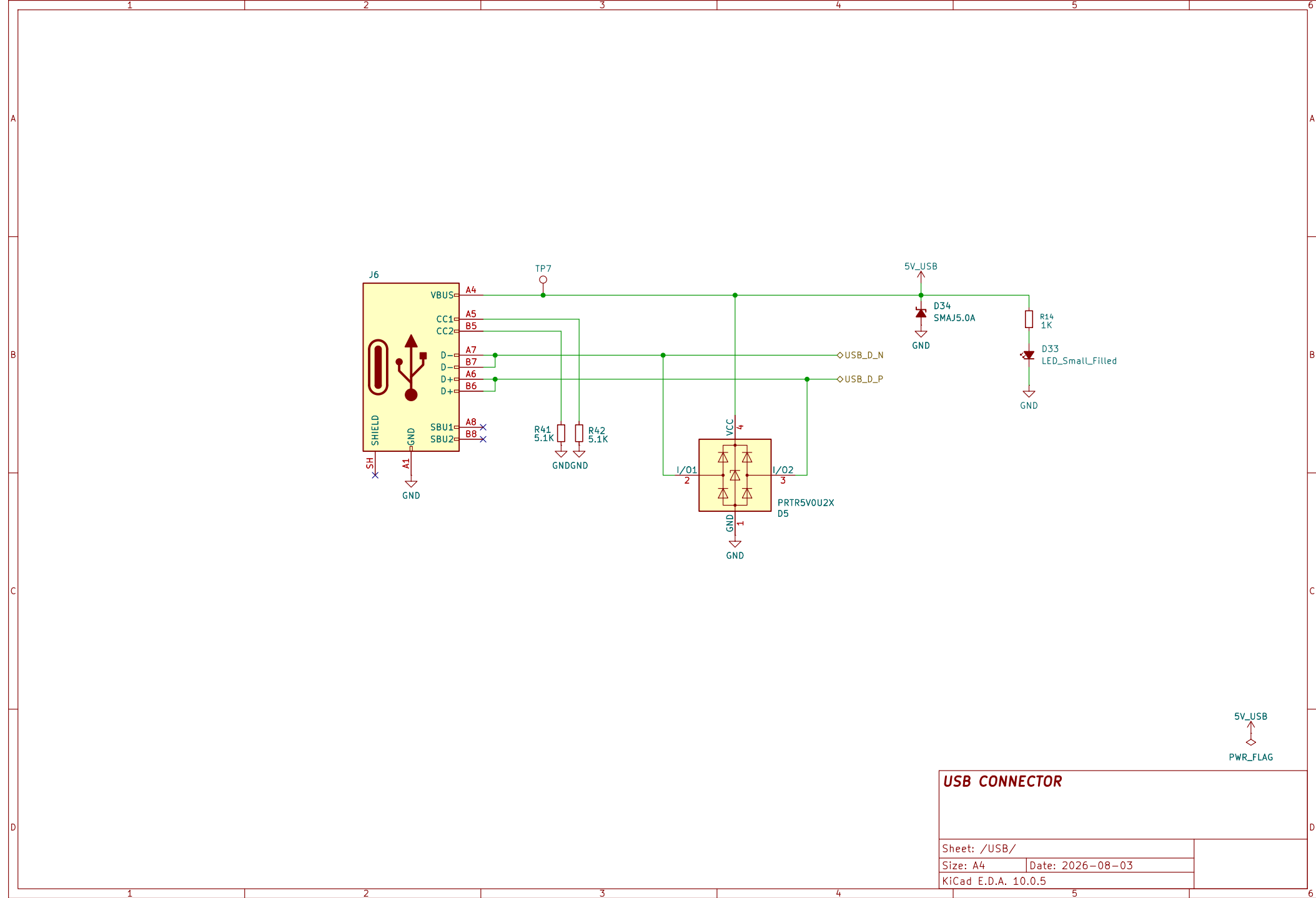
LEDs

Sheet: /MCU/leds2/

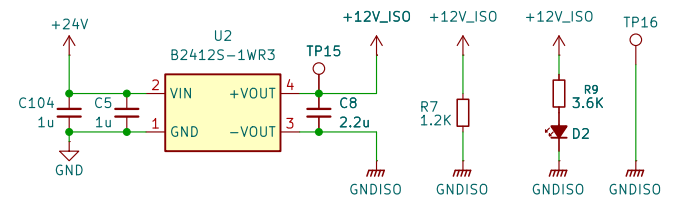
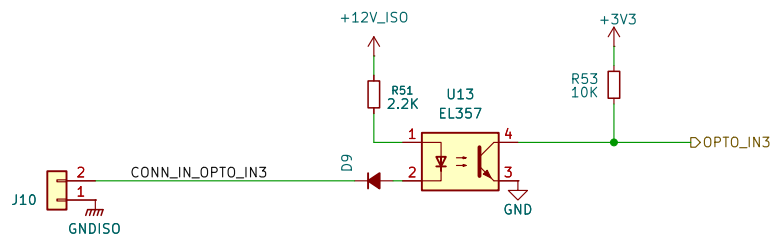
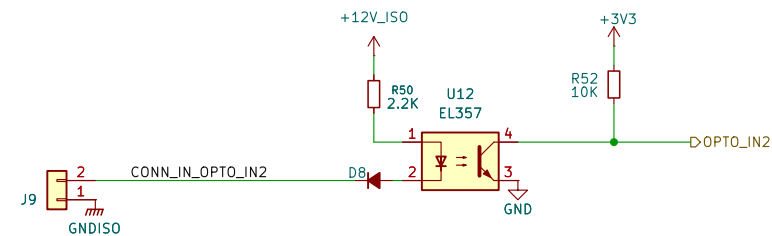
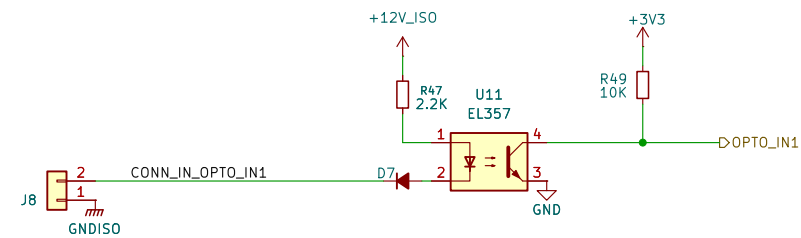
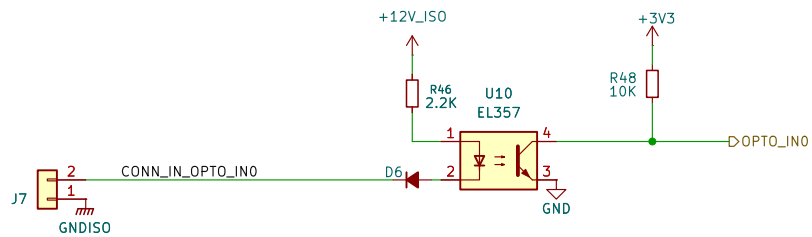
Size: A4 Date: 2026-08-03

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USB CONNECTOR		
Sheet: /USB/		
Size: A4	Date: 2026-08-03	
KiCad E.D.A. 10.0.5		



DESIGN NOTE: B2412S-1WR3 ISOLATED DC/DC

- Unregulated converter: Requires minimum 10% load (~8.3mA) to maintain stable 12V isolated output.
- Dummy Load (R50): 1.2kΩ, 1/4W minimum rating (draws 10mA).
- Power LED (D8 + R51): Draws ~2.8mA.
- Total Constant Load: ~12.8mA.

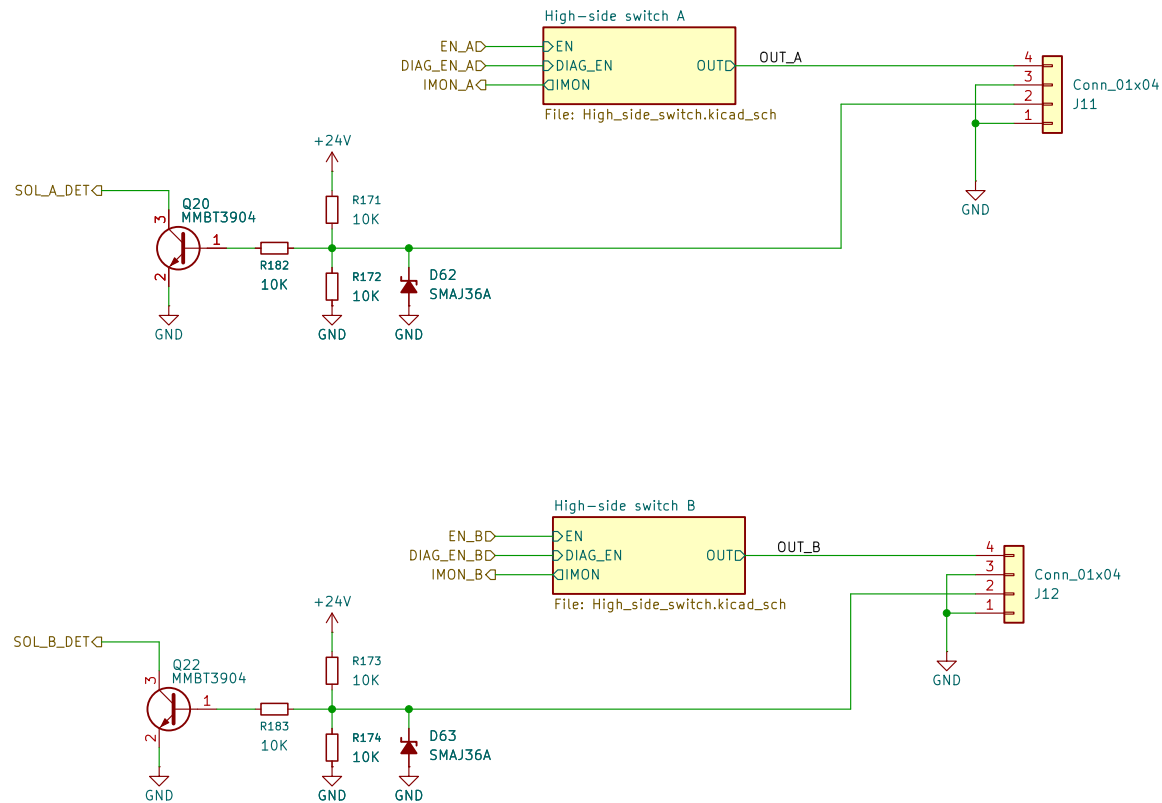
ISOLATED INPUTS

Sheet: /OPTO Inputs/

Size: A4

Date: 2026-08-03

KiCad E.D.A. 10.0.5



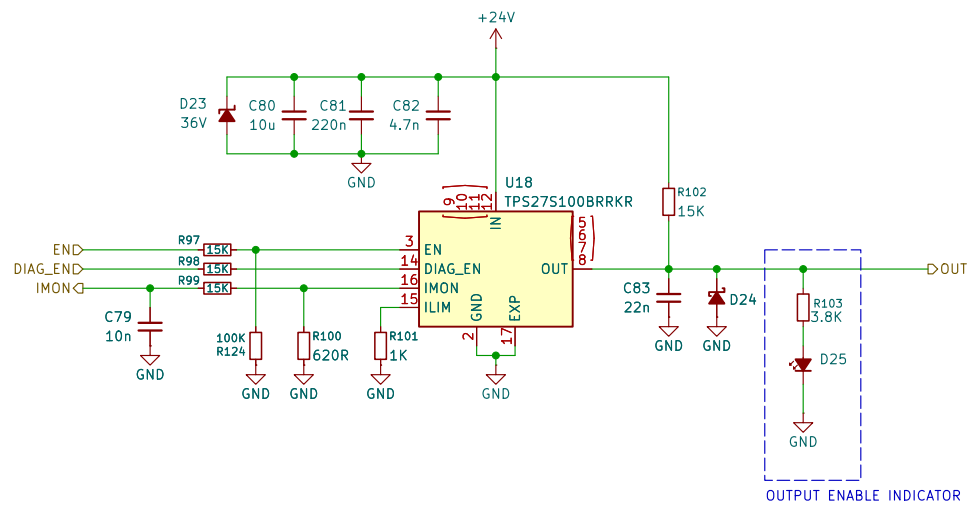
SOLENOID POWER OUTPUTS

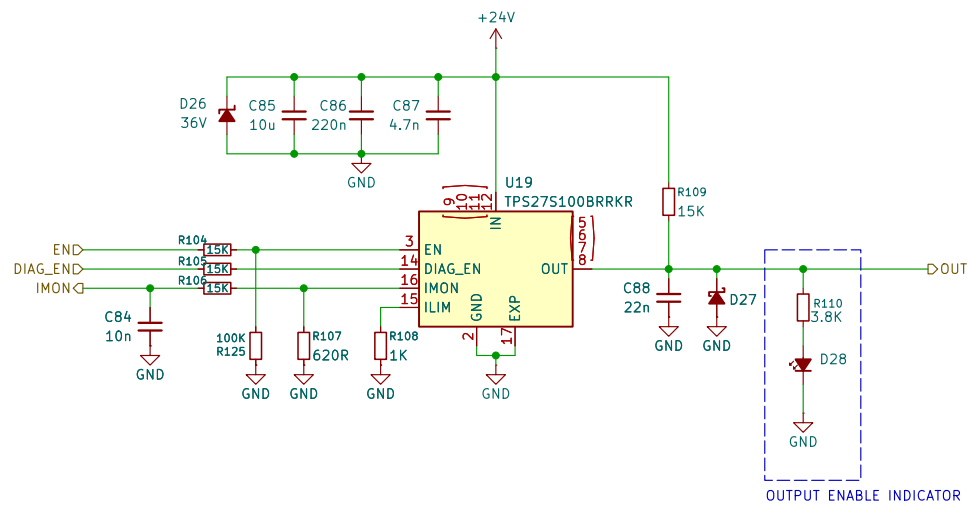
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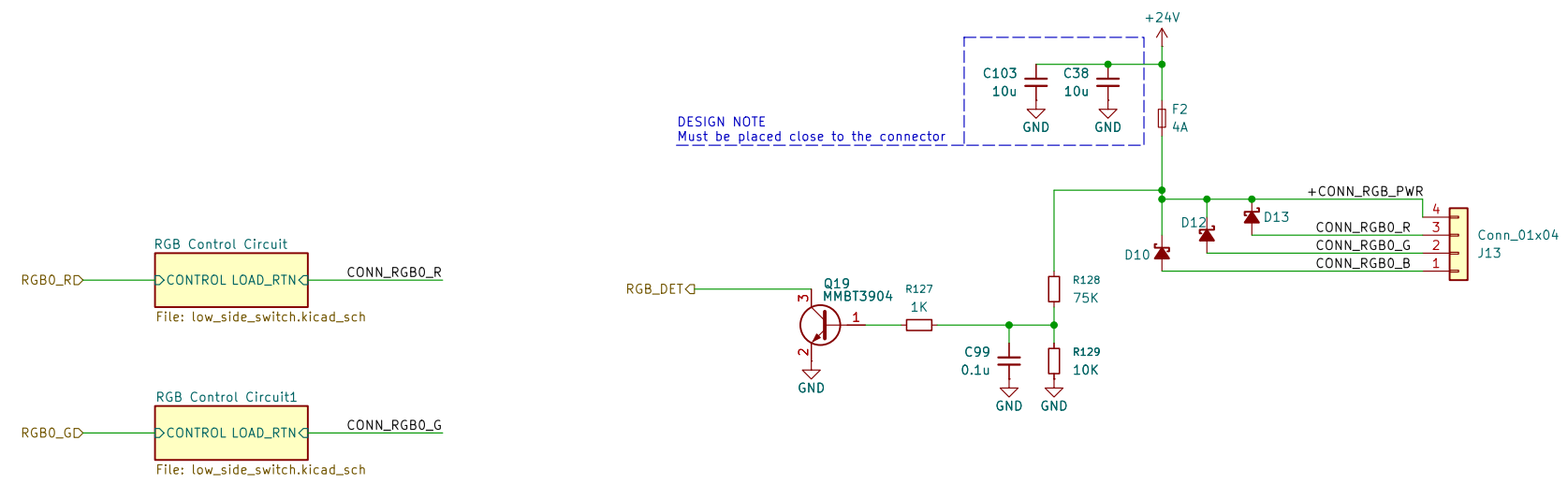
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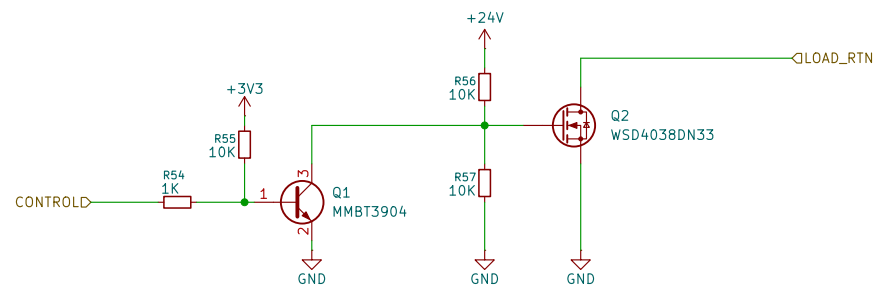
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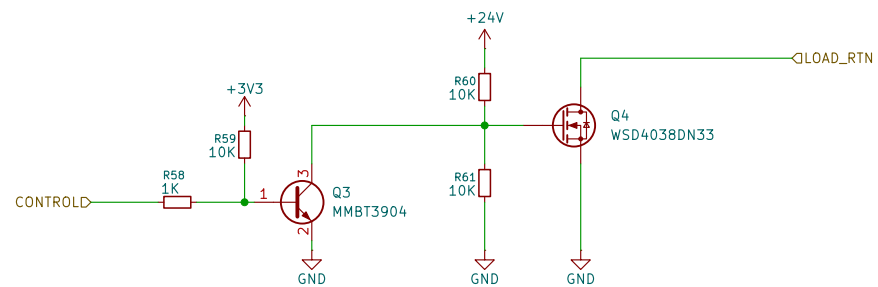




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Size: A4 Date: 2026-08-03

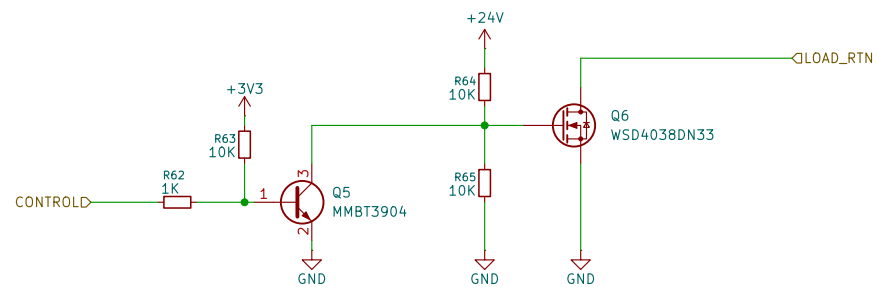
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Sheet: /RGB/RGB Control Circuit1/

Size: A4 Date: 2026-08-03

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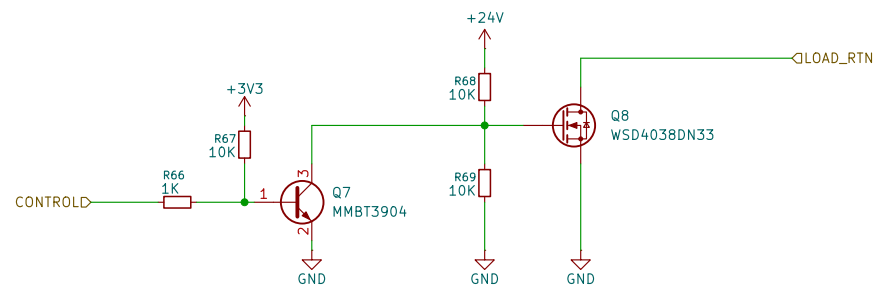
Sheet: /RGB/RGB Control Circuit2/

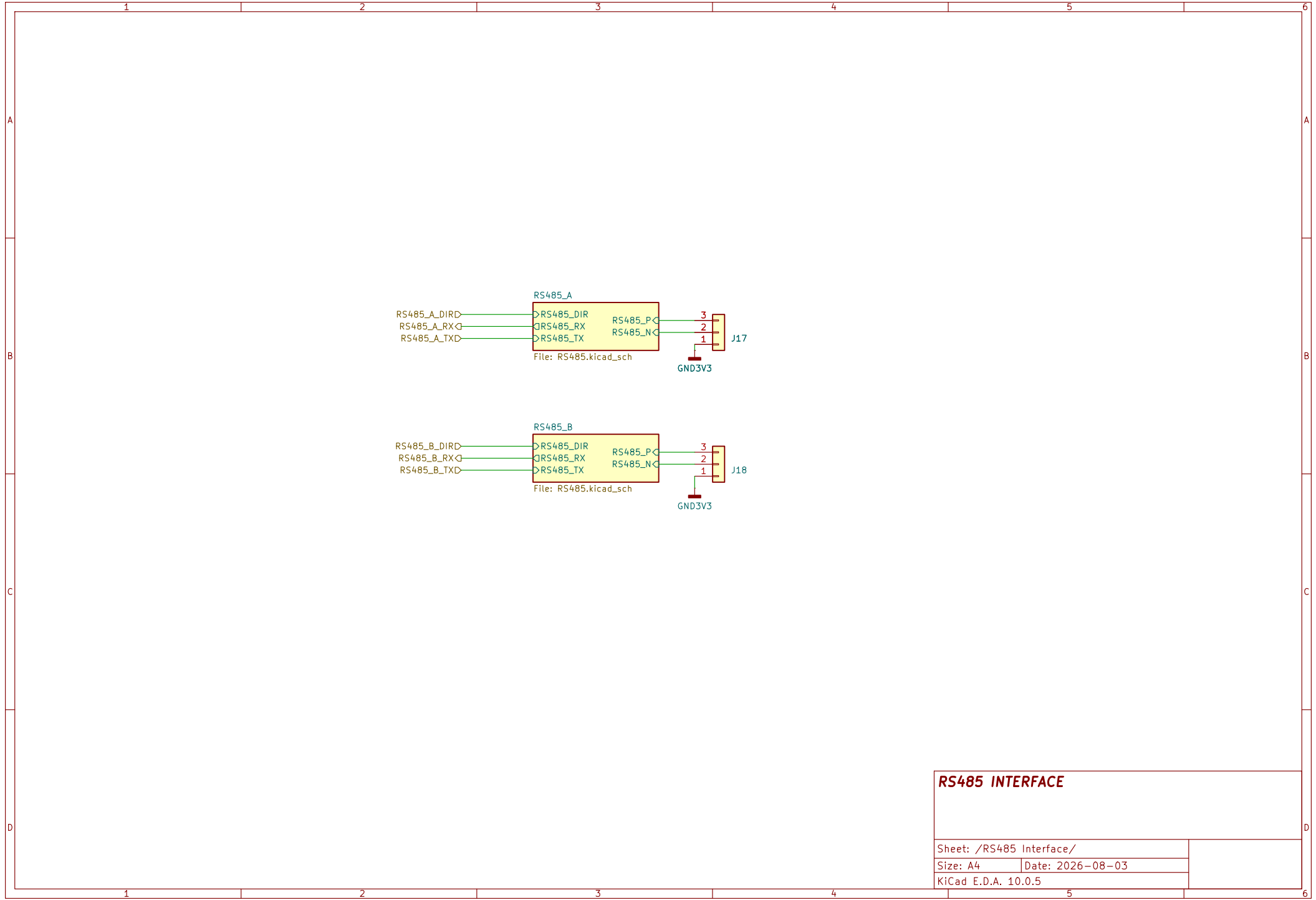
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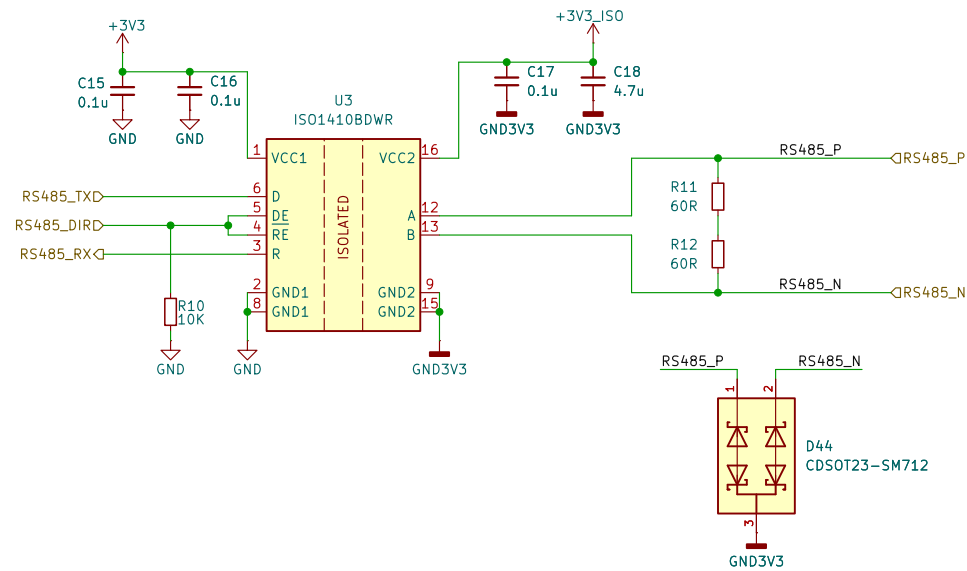


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RS485 INTERFACE		
Sheet: /RS485 Interface/		
Size: A4	Date: 2026-08-03	
KiCad E.D.A. 10.0.5		

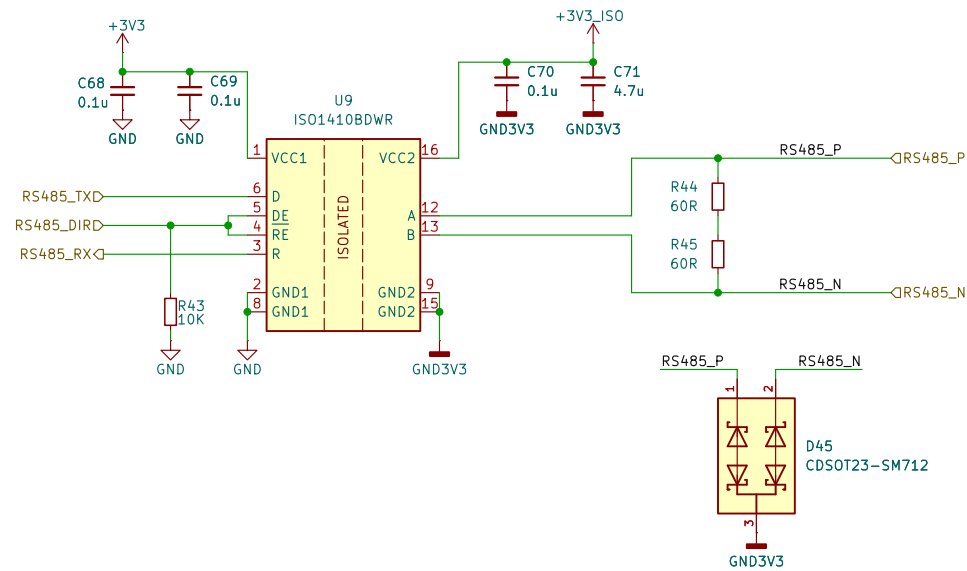


RS485 ISOLATED TRASCEIVER

Sheet: /RS485 Interface/RS485_A/

Size: A4 Date: 2026-08-03

KiCad E.D.A. 10.0.5

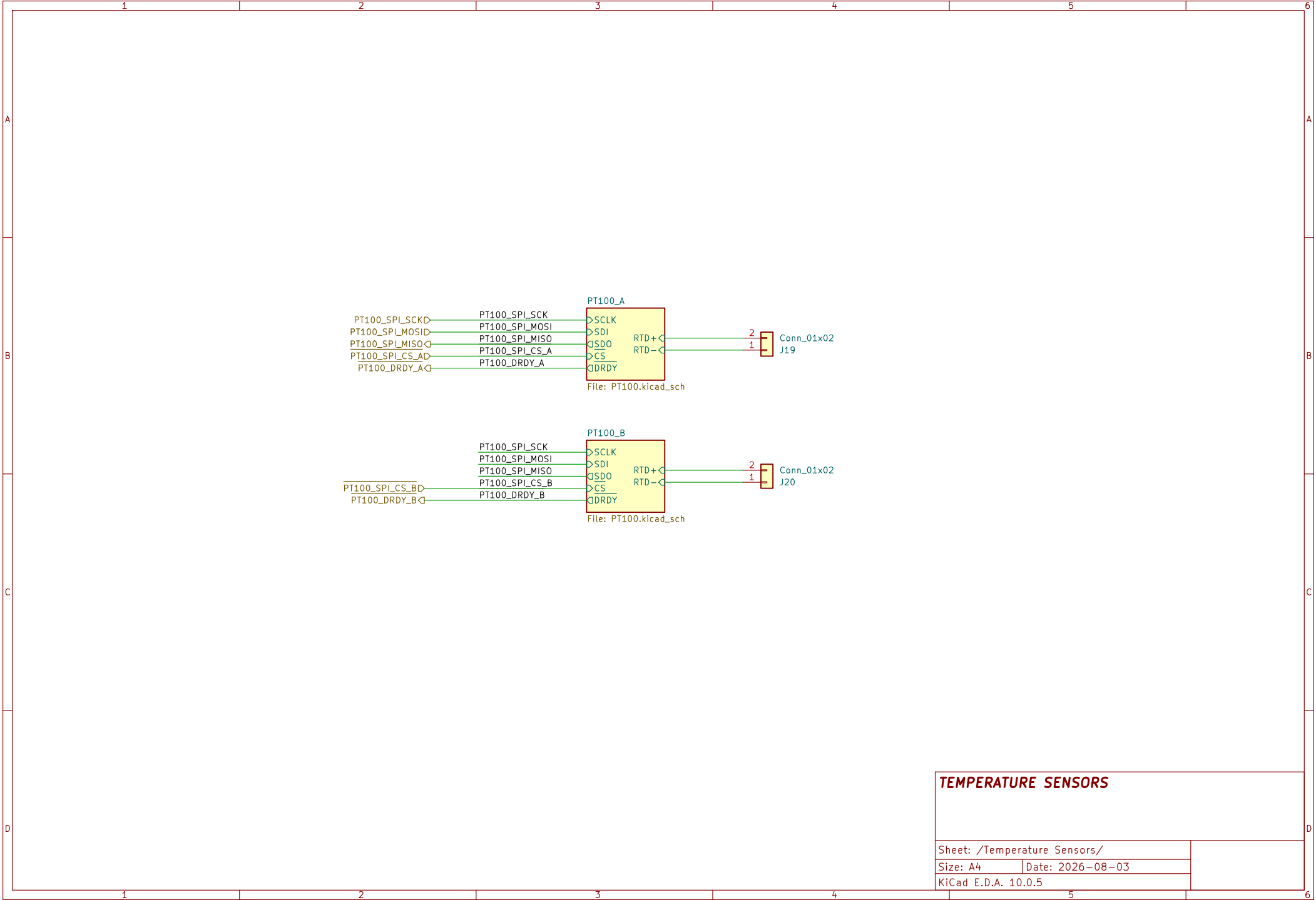


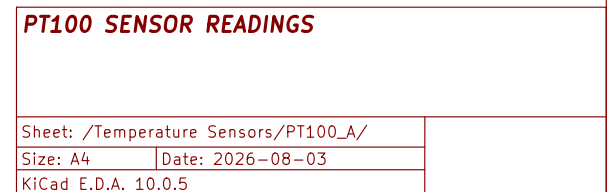
RS485 ISOLATED TRASCEIVER

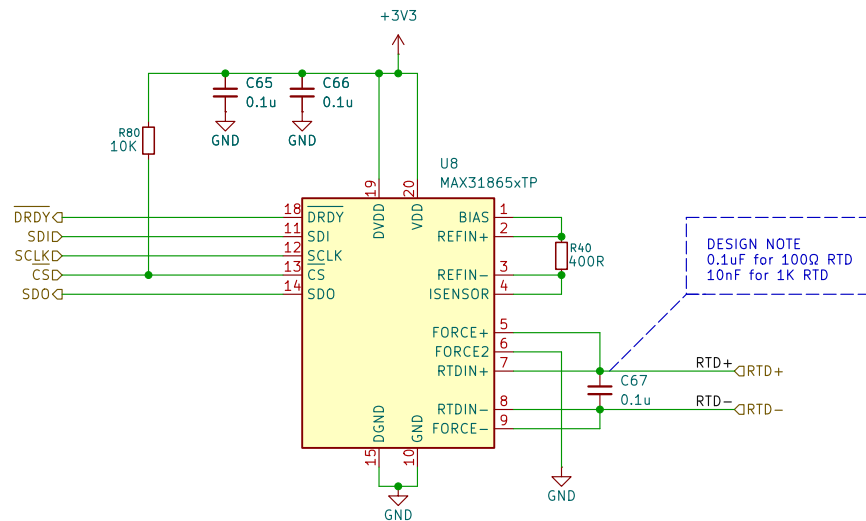
Sheet: /RS485 Interface/RS485_B/

Size: A4 Date: 2026-08-03

KiCad E.D.A. 10.0.5





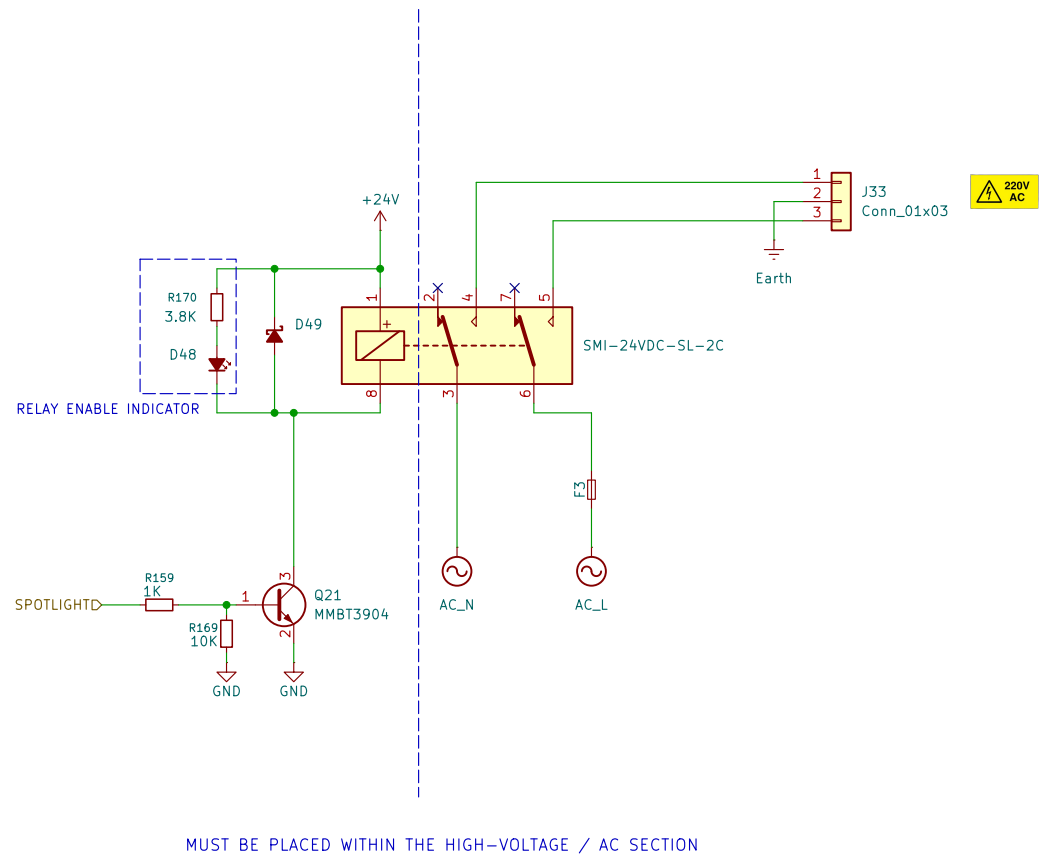


PT100 SENSOR READINGS

Sheet: /Temperature Sensors/PT100_B/

Size: A4 Date: 2026-08-03

KiCad E.D.A. 10.0.5

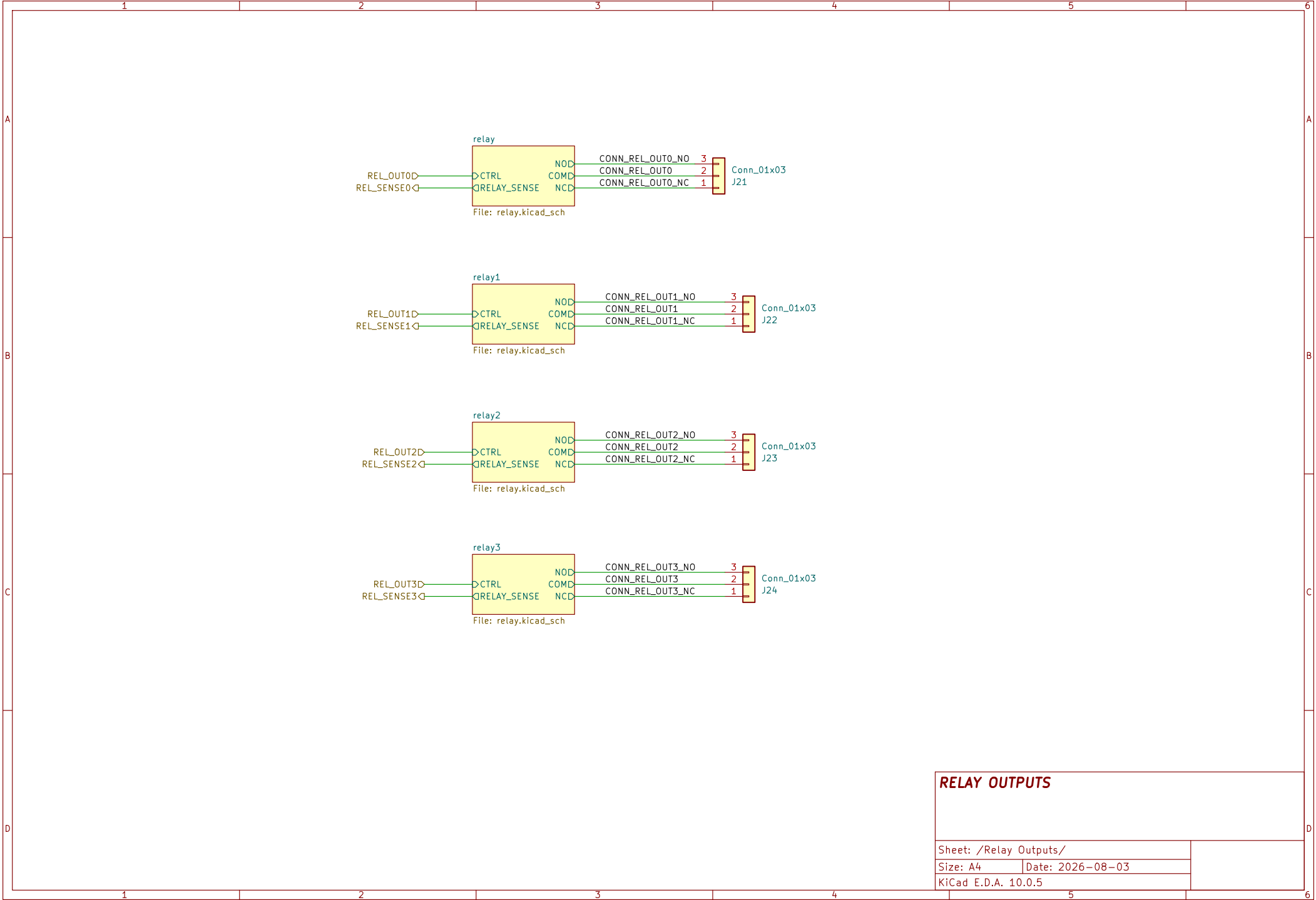


SPOTLIGHT

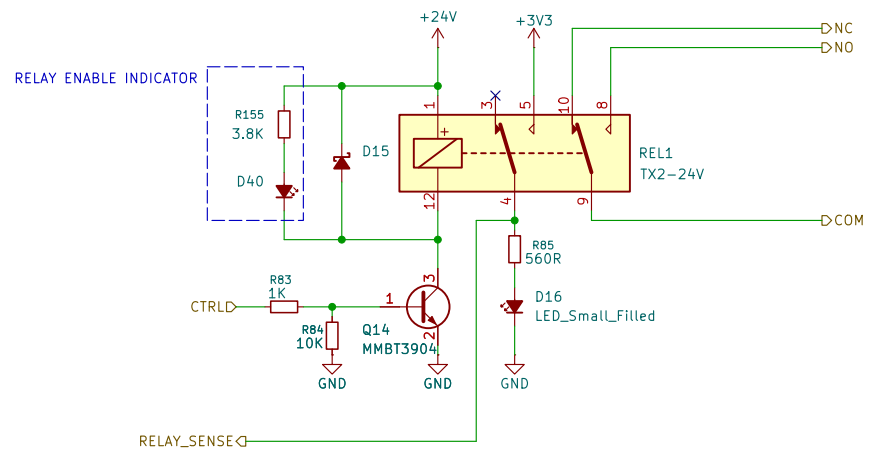
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Size: A4	Date: 2026-08-03
KiCad E.D.A. 10.0.5	

PWR_FLAG

Earth



RELAY OUTPUTS		
Sheet: /Relay Outputs/		
Size: A4	Date: 2026-08-03	
KiCad E.D.A. 10.0.5		

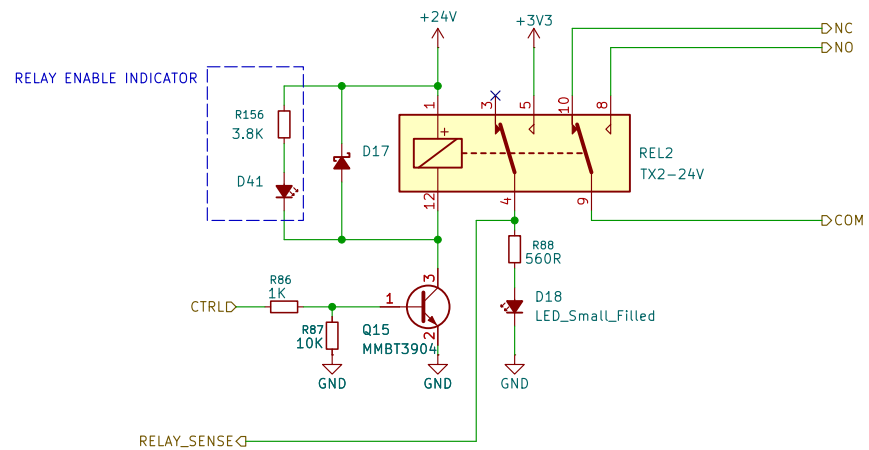


RELAY

Sheet: /Relay Outputs/relay/

Size: A4 Date: 2026-08-03

KiCad E.D.A. 10.0.5

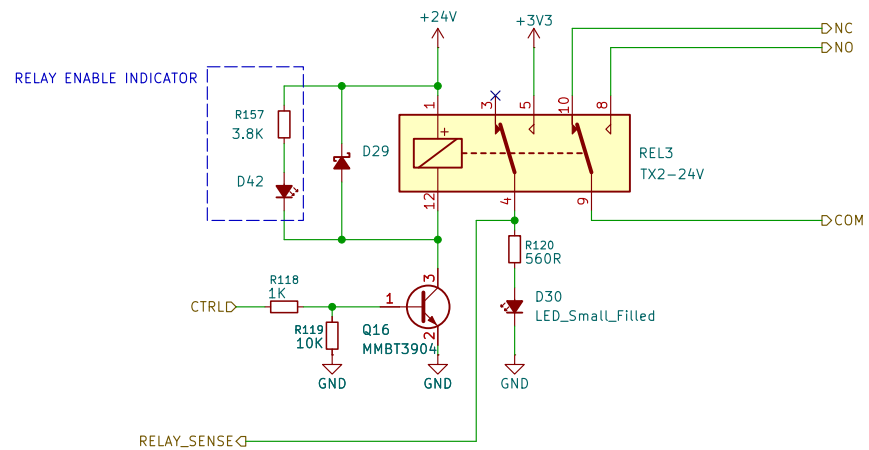


RELAY

Sheet: /Relay Outputs/relay1/

Size: A4 Date: 2026-08-03

KiCad E.D.A. 10.0.5

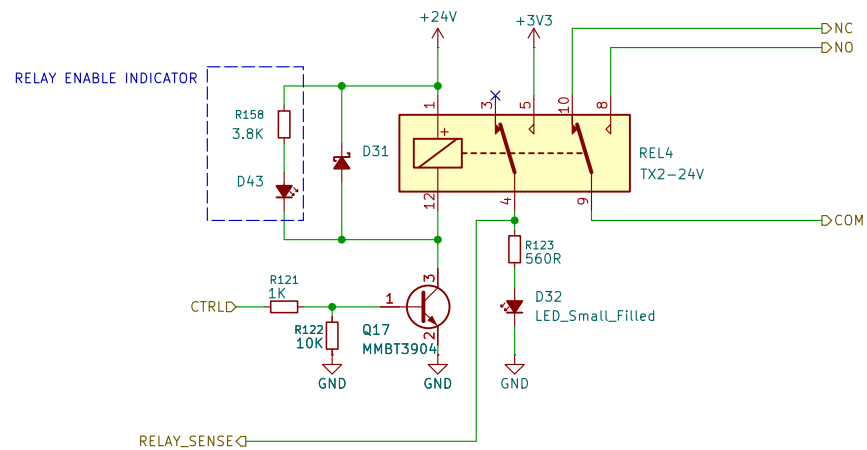


RELAY

Sheet: /Relay Outputs/relay2/

Size: A4 Date: 2026-08-03

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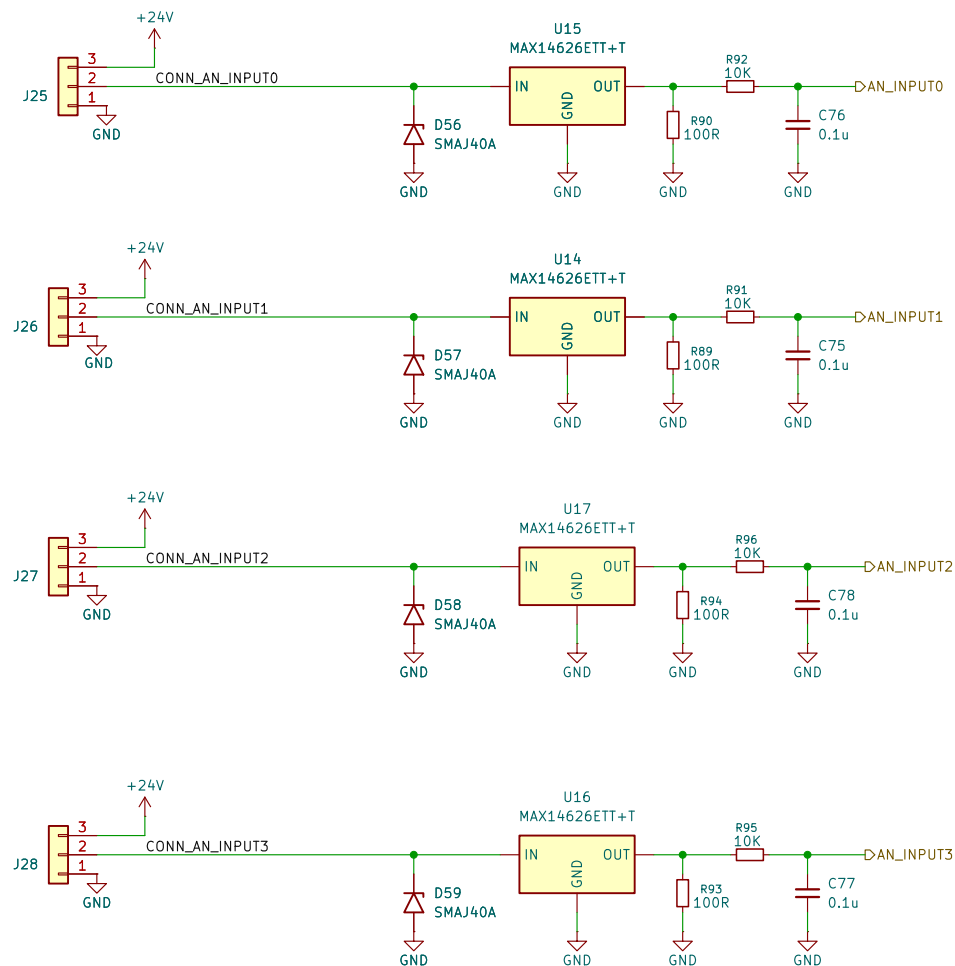


RELAY

Sheet: /Relay Outputs/relay3/

Size: A4 Date: 2026-08-03

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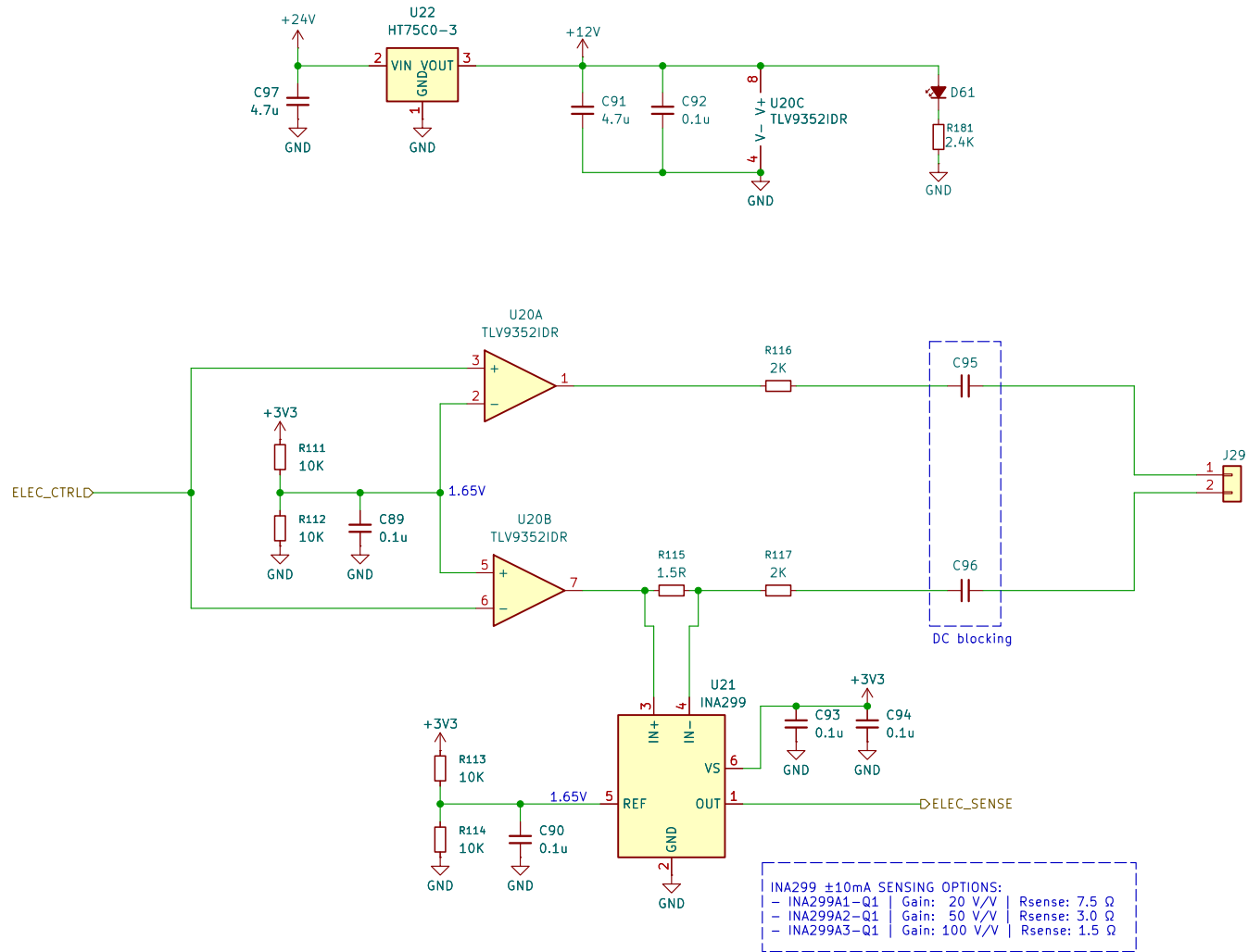


4-20mA INPUT PROTECTION

Sheet: /Analog Input/

Size: A4 Date: 2026-08-03

KiCad E.D.A. 10.0.5



ELECTRODE SENSE

Sheet: /Electrode/

Size: A4

Date: 2026-08-03

KiCad E.D.A. 10.0.5